

DAY 02 — EXCEL FORMULA FOUNDATION

1. Day 02 Objective

Build a strong Excel formula foundation for sales analysis, KPI calculations, data validation, and future SQL and Power BI work.

2. Topics Completed

- Formula structure and arithmetic operators
- Relative references
- Absolute references
- Mixed references
- SUM
- AVERAGE
- MIN and MAX
- COUNT and COUNTA
- Percentage calculations
- Achievement %
- Variance
- Growth %
- Combining formulas
- Formula error awareness
- Practical sales analysis
- Aggregation thinking

3. Formula Basics

Excel formulas normally begin with =.

Examples: =A2+B2 | =A2-B2 | =A2*B2 | =A2/B2 | =A2^2

Use cell references instead of hard-coded values whenever practical.

4. Relative, Absolute and Mixed References

Reference	Behavior	Memory
A1	Row and column move when copied	Both move
\$A\$1	Row and column stay fixed	Nothing moves
\$A1	Column fixed; row moves	Column stays
A\$1	Row moves; row 1 stays fixed	Row stays

Example: =B2*\$C\$2 → B2 changes when copied, while \$C\$2 stays fixed.

Example: =B2/B\$3 → the column can change while row 3 stays fixed.

5. Core Functions

SUM: =SUM(B2:B6) adds all numeric values in the range.

AVERAGE: =AVERAGE(B2:B6) calculates total ÷ number of values.

MIN: =MIN(B2:B6) returns the smallest numeric value.

MAX: =MAX(B2:B6) returns the largest numeric value.

COUNT: =COUNT(A1:A7) counts numeric cells only.

COUNTA: =COUNTA(A1:A7) counts all non-empty cells.

6. Percentage Metrics

Metric	Formula	Business Question
Achievement %	Actual ÷ Target	How much of target did we achieve?
Variance	Actual – Target	How far are we from target?
Growth %	(Current – Previous) ÷ Previous	How much did performance change?

Achievement example: $8,000 \div 10,000 = 80\%$.

Variance example: $8,000 - 10,000 = -2,000$.

Growth example: $10,000 \rightarrow 12,000 = 20\%$; $12,000 \rightarrow 9,000 = -25\%$.

Growth rule: always use the previous period as the denominator.

7. Combining Formulas

Example: =(B2-C2)/C2 combines subtraction and division to calculate the percentage difference between Sales and Target.

8. Formula Error Awareness

Error	Typical meaning	Example / note
#DIV/0!	Division by zero or blank denominator	=B2/C2 when C2 is 0
#VALUE!	Incompatible value/data type	Inappropriate text/number operation
#N/A	Value unavailable/not found	Common in lookup operations
#REF!	Invalid cell reference	Referenced cell/range became invalid

Analyst rule: an error is a signal to investigate, not something to blindly hide.

9. Practical Sales Dataset

Booking	Country	Salesperson	Sales	Target
B001	Tanzania	John	8,000	10,000
B002	Nigeria	David	12,000	10,000
B003	Kenya	Sarah	9,000	10,000

B004	Tanzania	Peter	11,000	10,000
B005	Nigeria	Mary	7,000	10,000
B006	Kenya	James	13,000	12,000

10. Practical Results

Analysis	Result
Total Sales	60,000
Total Target	62,000
Overall Achievement	96.77%
Total Variance	-2,000

11. Country-Level Aggregation

Country	Sales	Target	Variance	Achievement
Tanzania	19,000	20,000	-1,000	95%
Nigeria	19,000	20,000	-1,000	95%
Kenya	22,000	22,000	0	100%
Total	60,000	62,000	-2,000	96.77%

12. Critical Analyst Lesson: Row vs Aggregation

B002 individually achieved 120%, but Nigeria's two bookings together achieved 95%. A row-level result must not be treated as the country-level result.

Investigation path: Booking → Country → Branch → Salesperson → Product/Service → Time, plus Customer, Booking Status or other relevant dimensions when needed.

13. Day 02 Golden Rules

- Use cell references instead of hard-coding values whenever practical.
- Know exactly what moves and what stays fixed in a formula.
- Choose the function according to the question being answered.
- Do not confuse Achievement %, Variance and Growth %.
- Use the previous period as the denominator for growth.
- Investigate formula errors instead of simply hiding them.
- Validate calculations before interpreting them.
- Never confuse row-level performance with aggregated performance.
- Always understand the grain/level at which a conclusion is being made.

14. Day 02 Checklist

- ☒ Formula structure
- ☒ Arithmetic operators
- ☒ Relative references
- ☒ Absolute references
- ☒ Mixed references
- ☒ SUM
- ☒ AVERAGE
- ☒ MIN
- ☒ MAX
- ☒ COUNT
- ☒ COUNTA
- ☒ Percentage calculations
- ☒ Achievement %
- ☒ Variance
- ☒ Growth %
- ☒ Combining formulas
- ☒ Formula error awareness
- ☒ Practical sales analysis
- ☒ Aggregation thinking